

Notebook Computer

P775TM (-G) / P775TM1 (-G)

Service Manual

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About this Manual

This manual is intended for service personnel who have completed sufficient training to undertake the maintenance and inspection of personal computers.

It is organized to allow you to look up basic information for servicing and/or upgrading components of the ***P775TM (-G) / P775TM1 (-G)*** series notebook PC.

The following information is included:

Chapter 1, Introduction, provides general information about the location of system elements and their specifications.

Chapter 2, Disassembly, provides step-by-step instructions for disassembling parts and subsystems and how to upgrade elements of the system.

Appendix A, Part Lists

Appendix B, Schematic Diagrams

Appendix C, Updating the FLASH ROM BIOS

IMPORTANT SAFETY INSTRUCTIONS

Follow basic safety precautions, including those listed below, to reduce the risk of fire, electric shock and injury to persons when using any electrical equipment:

1. Do not use this product near water, for example near a bath tub, wash bowl, kitchen sink or laundry tub, in a wet basement or near a swimming pool.
2. Avoid using a telephone (other than a cordless type) during an electrical storm. There may be a remote risk of electrical shock from lightning.
3. Do not use the telephone to report a gas leak in the vicinity of the leak.
4. Use only the power cord and batteries indicated in this manual. Do not dispose of batteries in a fire. They may explode. Check with local codes for possible special disposal instructions.
5. This product is intended to be supplied by a Listed Power Unit with an AC Input of 100 - 240V, 50 - 60Hz, DC Output of 19.5V, 16.9A (**330** Watts) or 16.9A (**330** Watts) minimum AC/DC Adapter.

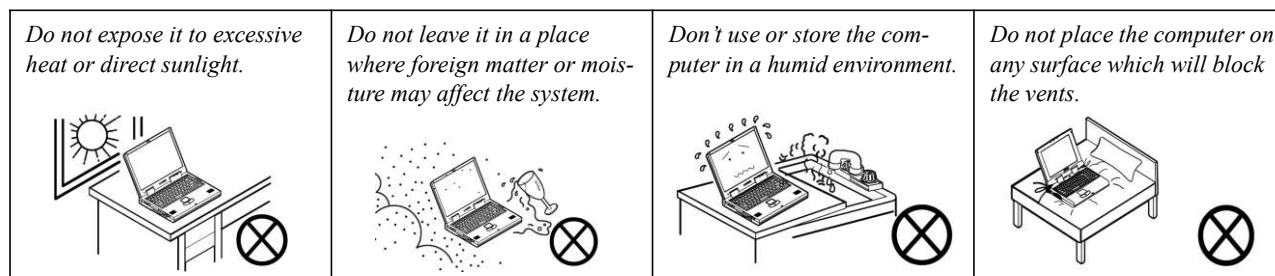
Instructions for Care and Operation

The notebook computer is quite rugged, but it can be damaged. To prevent this, follow these suggestions:

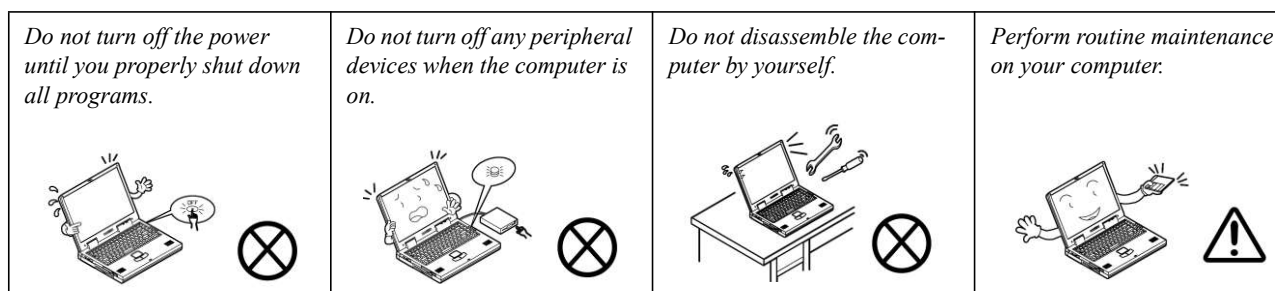
1. **Don't drop it, or expose it to shock.** If the computer falls, the case and the components could be damaged.



2. **Keep it dry, and don't overheat it.** Keep the computer and power supply away from any kind of heating element. This is an electrical appliance. If water or any other liquid gets into it, the computer could be badly damaged.



3. **Follow the proper working procedures for the computer.** Shut the computer down properly and don't forget to save your work. Remember to periodically save your data as data may be lost if the battery is depleted.



Preface



Removal Warning

When removing any cover(s) and screw(s) for the purposes of device upgrade, remember to replace the cover(s) and screw(s) before restoring power to the system.

Also note the following when the cover is removed:

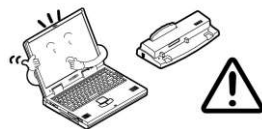
- Hazardous moving parts.
- Keep away from moving fan blades

Power Safety Warning

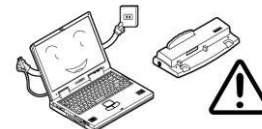
Before you undertake any upgrade procedures, make sure that you have turned off the power, and disconnected all peripherals and cables (including telephone lines and power cord). You must also remove your battery in order to prevent accidentally turning the machine on.

4. **Avoid interference.** Keep the computer away from high capacity transformers, electric motors, and other strong magnetic fields. These can hinder proper performance and damage your data.
5. **Take care when using peripheral devices.**

Use only approved brands of peripherals.



Unplug the power cord before attaching peripheral devices.



Power Safety

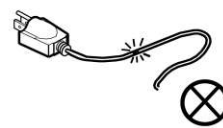
The computer has specific power requirements:

- Only use a power adapter approved for use with this computer.
- Your AC adapter may be designed for international travel but it still requires a steady, uninterrupted power supply. If you are unsure of your local power specifications, consult your service representative or local power company.
- The power adapter may have either a 2-prong or a 3-prong grounded plug. The third prong is an important safety feature; do not defeat its purpose. If you do not have access to a compatible outlet, have a qualified electrician install one.
- When you want to unplug the power cord, be sure to disconnect it by the plug head, not by its wire.
- Make sure the socket and any extension cord(s) you use can support the total current load of all the connected devices.
- Before cleaning the computer, make sure it is disconnected from any external power supplies.

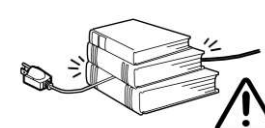
Do not plug in the power cord if you are wet.



Do not use the power cord if it is broken.



Do not place heavy objects on the power cord.



Battery Precautions

- Only use batteries designed for this computer. The wrong battery type may explode, leak or damage the computer.
- Do not continue to use a battery that has been dropped, or that appears damaged (e.g. bent or twisted) in any way. Even if the computer continues to work with a damaged battery in place, it may cause circuit damage, which may possibly result in fire.
- Recharge the batteries using the notebook's system. Incorrect recharging may make the battery explode.
- Do not try to repair a battery pack. Refer any battery pack repair or replacement to your service representative or qualified service personnel.
- Keep children away from, and promptly dispose of a damaged battery. Always dispose of batteries carefully. Batteries may explode or leak if exposed to fire, or improperly handled or discarded.
- Keep the battery away from metal appliances.
- Affix tape to the battery contacts before disposing of the battery.
- Do not touch the battery contacts with your hands or metal objects.

Battery Guidelines

The following can also apply to any backup batteries you may have.

- If you do not use the battery for an extended period, then remove the battery from the computer for storage.
- Before removing the battery for storage charge it to 60% - 70%.
- Check stored batteries at least every 3 months and charge them to 60% - 70%.



Battery Disposal

The product that you have purchased contains a rechargeable battery. The battery is recyclable. At the end of its useful life, under various state and local laws, it may be illegal to dispose of this battery into the municipal waste stream. Check with your local solid waste officials for details in your area for recycling options or proper disposal.

Caution

Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Discard used battery according to the manufacturer's instructions.

Battery Level

Click the battery icon  in the taskbar to see the current battery level and charge status. A battery that drops below a level of 10% will not allow the computer to boot up. Make sure that any battery that drops below 10% is recharged within one week.

Preface

Related Documents

You may also need to consult the following manual for additional information:

User's Manual on Disc

This describes the notebook PC's features and the procedures for operating the computer and its ROM-based setup program. It also describes the installation and operation of the utility programs provided with the notebook PC.

System Startup

1. Remove all packing materials.
2. Place the computer on a stable surface.
3. Insert the battery and make sure it is locked in position.
4. Securely attach any peripherals you want to use with the computer (e.g. keyboard and mouse) to their ports.
5. **When first setting up the computer use the following procedure** (as to safeguard the computer during shipping, the battery will be locked to not power the system until first connected to the AC/DC adapter and initially set up as below):
 - Attach the AC/DC adapter cord to the DC-In jack at the rear of the computer, then plug the AC power cord into an outlet, and connect the AC power cord to the AC/DC adapter and **leave it there for 6 seconds or longer**.
 - Remove the adapter cord from the computer's DC-In jack, and then plug it back in again; the battery will now be unlocked.
6. Use one hand to raise the lid/LCD to a comfortable viewing angle (do not to exceed **135** degrees); use the other hand (as illustrated in [Figure 1](#)) to support the base of the computer (**Note: Never** lift the computer by the lid/LCD).
7. Press the power button to turn the computer "on".

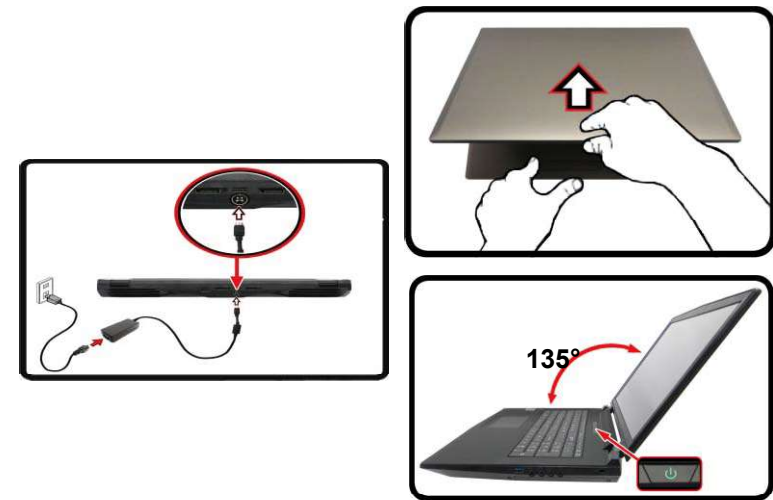


Figure 1
**Opening the Lid/LCD/
Computer with AC/DC
Adapter Plugged-In**

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Chapter 1: Introduction

Overview

This manual covers the information you need to service or upgrade the **P775TM (-G) / P775TMI (-G)** series notebook computer. Information about operating the computer (e.g. getting started, and the *Setup* utility) is in the *User's Manual*. Information about drivers (e.g. VGA & audio) is also found in *User's Manual*. That manual is shipped with the computer.

Operating systems (e.g. *Windows 10*, etc.) have their own manuals as do application software (e.g. word processing and database programs). If you have questions about those programs, you should consult those manuals.

The **P775TM (-G) / P775TMI (-G)** series notebook is designed to be upgradeable. See [Disassembly on page 2 - 1](#) for a detailed description of the upgrade procedures for each specific component. Please note the warning and safety information indicated by the “” symbol.

The balance of this chapter reviews the computer's technical specifications and features.

Introduction

Specifications



Latest Specification Information

The specifications listed here are correct at the time of sending them to the press. Certain items (particularly processor types/speeds) may be changed, delayed or updated due to the manufacturer's release schedule. Check with your service center for more details.



CPU Speed & Computer in DC Mode

Note that when the computer is in DC mode (powered by the battery only) the CPU may not run at full speed. This is a design feature implemented in order to protect the battery.

Processor Options

i7-8700K (3.70GHz)*

12MB Smart Cache, **14nm**, DDR4-2666MHz, TDP 95W

i5-8600K (3.60GHz)*

9MB Smart Cache, **14nm**, DDR4-2666MHz, TDP 95W

i7-8700 (3.20GHz)

12MB Smart Cache, **14nm**, DDR4-2666MHz, TDP 65W

i5-8400 (2.80GHz)

6MB Smart Cache, **14nm**, DDR4-2666MHz, TDP 65W

i3-8350K (4.00GHz) (for P775TM/P775TM1 only)

8MB Smart Cache, **14nm**, DDR4-2400MHz, TDP 91W

i3-8100 (3.60GHz) (for P775TM/P775TM1 only)

6MB Smart Cache, **14nm**, DDR4-2400MHz, TDP 65W

**Supports Intel® XTU overclocking technology depending on CPU SKU*

LCD Options

17.3" (43.94cm), 16:9, QFHD (3840x2160)/FHD (1920x1080)

Core Logic

Intel® Z370 Chipset

BIOS

AMI BIOS (64Mb SPI Flash-ROM)

Pointing Device

Built-In Secure Pad (with Microsoft PTP Multi Gesture & Scrolling Functionality)

Keyboard

Full Color **Illuminated** Full-size Winkey Keyboard (with numeric keypad and anti-ghost keys)

Video Adapter Options

NVIDIA® GeForce GTX 1060 PCIe Video Card

6GB GDDR5 Video RAM on board

Supports GPU Overclocking

NVIDIA® GeForce GTX 1070 / GTX 1080 PCIe Video Card (for P775TM1(-G) only)

8GB GDDR5 Video RAM

Supports GPU Overclocking

Memory

Four 260 Pin SO-DIMM Sockets Supporting **DDR4 2400 MHz** Memory

(The real memory operating frequency depends on the FSB of the processor.)

Memory Expandable from **8GB (minimum)** up to **64GB (maximum)**

Support XMP 2666/3000 MHz (XMP depends on processor)



SO-DIMM Memory Types

All SO-DIMM memory modules installed in the system should be identical (the same size and brand) in order to prevent unexpected system behavior.

Do not mix SO-DIMM memory module sizes and brands otherwise unexpected system problems may occur.

Security

Security (Kensington® Type) Lock Slot

BIOS Password

Fingerprint Reader Module

Trusted Platform Module 2.0

Intel PTT for Systems Without TPM Hardware

Storage

Two changeable 2.5" (6cm) 7.0mm (h)/ 9.5mm (h) SATA (Serial) Hard Disk Drives/Solid State Drives (SSD) supporting RAID level 0/1

(Factory Option) Two M.2 **SATA** 2280 SSDs supporting RAID level 0/1

Or

(Factory Option) Two M.2 **PCIe**

Gen3 x4 2280 SSDs supporting RAID level 0/1

Audio

High Definition Audio Compliant Interface
S/PDIF Digital Output
Two Speakers
Sound Blaster Audio
ESS™ SABRE HIFI DAC for High Resolution Headphone Audio
Built-In Array Microphone
Sub-Woofer

Note: External 7.1CH Audio Output Supported by 2-In-1 Audio Jacks, Microphone, Line-In and Line- Out Jacks

Interface

One USB 3.1 Gen 2 Type C Port
One USB 3.1 Gen 2/Thunderbolt 3 Combo Port (Type C)
Two Mini DisplayPorts (1.3)
One HDMI-Out Port
One 2-In-1 Audio Jack (Headphone/ S/PDIF Optical Output Combo Jack)
One Microphone-In Jack
One Line-Out Jack
One Line-In Jack
One RJ-45 LAN Jacks
One DC-In Jack
Four USB 3.0 (USB 3.1 Gen 1) Ports (Including one AC/DC Powered USB port)

Communication

Built-In Qualcomm 10/100/1000Mb Base-TX Ethernet LAN
2.0M FHD PC Camera Module

WLAN/ Bluetooth M.2 Modules:

(**Factory Option**) Intel® Wireless-AC 8265 Wireless LAN (802.11ac) + Bluetooth 4.1

(**Factory Option**) Qualcomm® Atheros Killer™ Wireless-AC 1535 Wireless LAN (802.11ac) + Bluetooth 4.1

(**Factory Option**) Qualcomm® Wireless LAN (802.11ac/ad) + Bluetooth 4.1

M.2 Slots

Slot 1 for **Combo WLAN and Bluetooth** Module

Slot 2 for **SATA** or **PCIe Gen3 x4 SSD**

Slot 3 for **SATA** or **PCIe Gen3 x4 SSD**

Card Reader

Embedded Multi-In-1 Push-Push Card Reader
MMC (MultiMedia Card) / RS MMC
SD (Secure Digital) / Mini SD / SDHC/ SDXC (up to UHS-II)

Features

Supports NVIDIA® G-SYNC™ Technology
(NVIDIA® G-SYNC™ Technology is supported by some LCD panels and GTX series video adapters Only)
Intel® Optane™ Technology
Virtual Reality Ready (*i5 & i7 Processor only*)
Supports Windows® 10 Cortana with Voice
(**Factory Option**) USB Drive

Environmental Spec

Temperature

Operating: 10°C - 35°C

Non-Operating: -20°C - 60°C

Relative Humidity

Operating: 20% - 80%

Non-Operating: 10% - 90%

Power

Removable 8-cell Smart Lithium-Ion Battery Pack, 82WH

Full Range AC/DC Adapter

AC Input: 100 - 240V, 50 - 60Hz

DC Output: 19.5V, 11.8A (**230W**) - P775TM(-G)

DC Output: 19.5V, 16.9A (**330W**) - P775TM1(-G)

Dimensions & Weight

418mm (w) * 295.3mm (d) * 40.9mm (h)

3.9kg (Barebone System with Video Card and 82WH Battery)

Or

4.3kg (Barebone System with Video Card and 82WH Battery)

Introduction

Figure 1
Top View

1. PC Camera
2. PC Camera LED
3. Built-In Microphone
4. LCD
5. Speakers
6. Power Button
7. LED Lock Indicators
8. Keyboard
9. TouchPad and Buttons
10. Fingerprint Reader

External Locator - Top View with LCD Panel Open



External Locator - Front & Right side Views

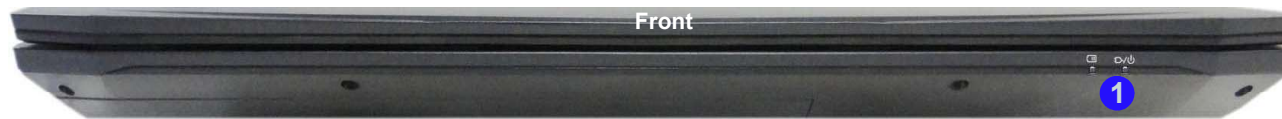


Figure 2
Front Views

1. LED Power Indicators



Figure 3
Right Side Views

1. Line-In Jack
2. Microphone Jack
3. Line-Out Jack
4. Headphone and S/PDIF Combo Jack
5. USB 3.0/3.1 Port
6. Security Lock Slot

Introduction

External Locator - Left Side & Rear View

Figure 4
Left Side View

1. RJ-45 LAN Jack
2. USB 3.1/
Thunderbolt
Combo Port
3. USB 3.1 Port
4. USB 3.0/3.1 Port
5. Powered USB 3.1
Port
6. Multi-in-1 Card
Reader

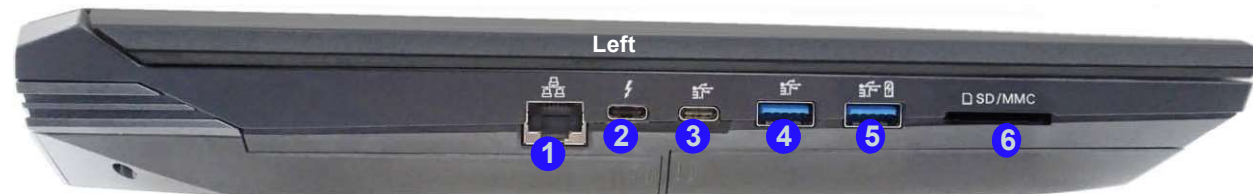


Figure 5
Rear View

1. Vent/Fan Intake
2. HDMI-Out Port
3. Mini Display Port 1
4. Mini Display Port 2
5. DC-In Jack



External Locator - Bottom View



Figure 6
Bottom View

1. Vent
2. Component Bay Cover
3. Battery
4. Sub Woofer
5. HDD Bay



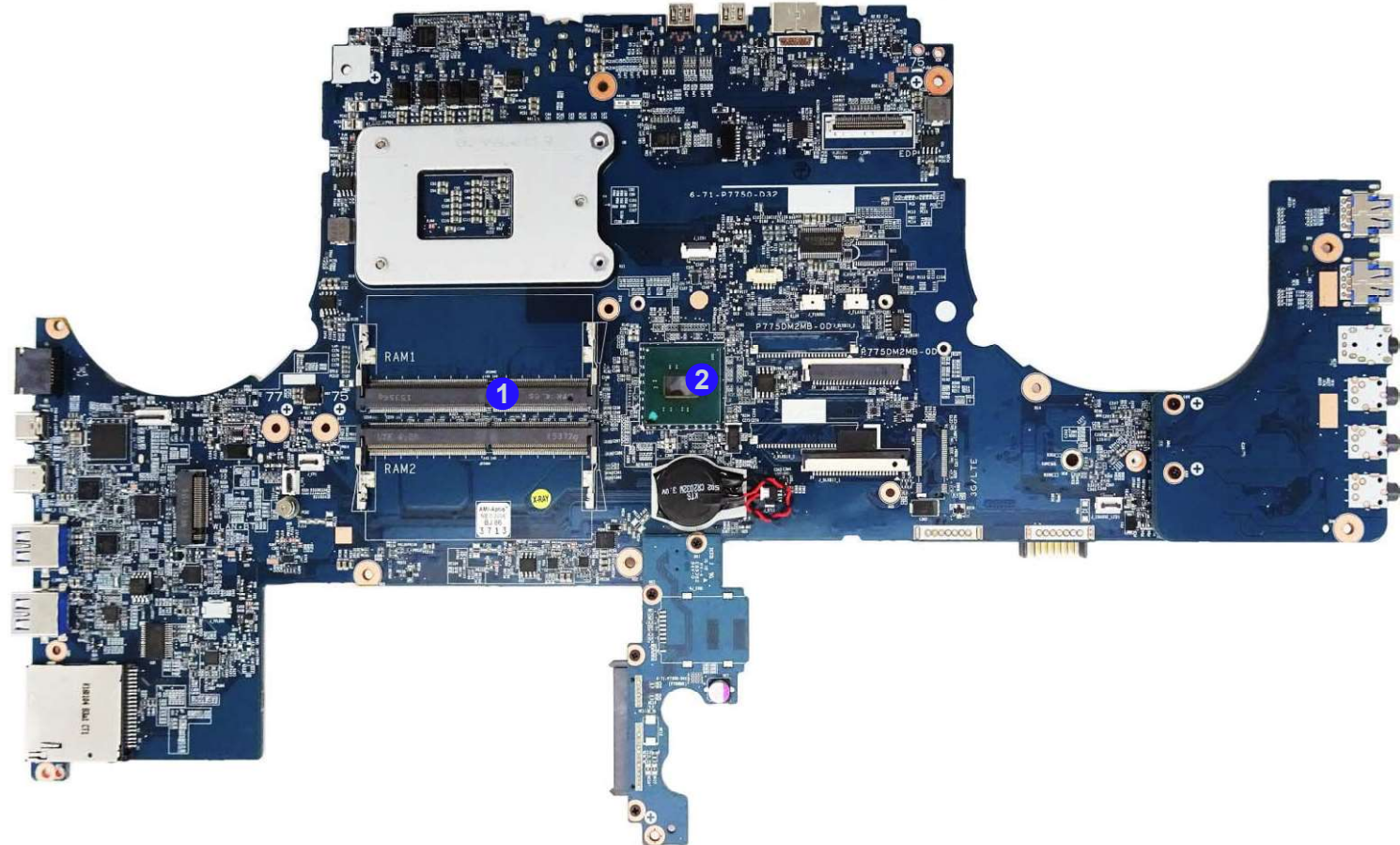
Overheating

To prevent your computer from overheating make sure nothing blocks the vent/fan intakes while the computer is in use.

Figure 7
**Mainboard Top
Key Parts**

1. Memory Slots
DDR4 SO-DIMM
2. Platform
Controller Hub

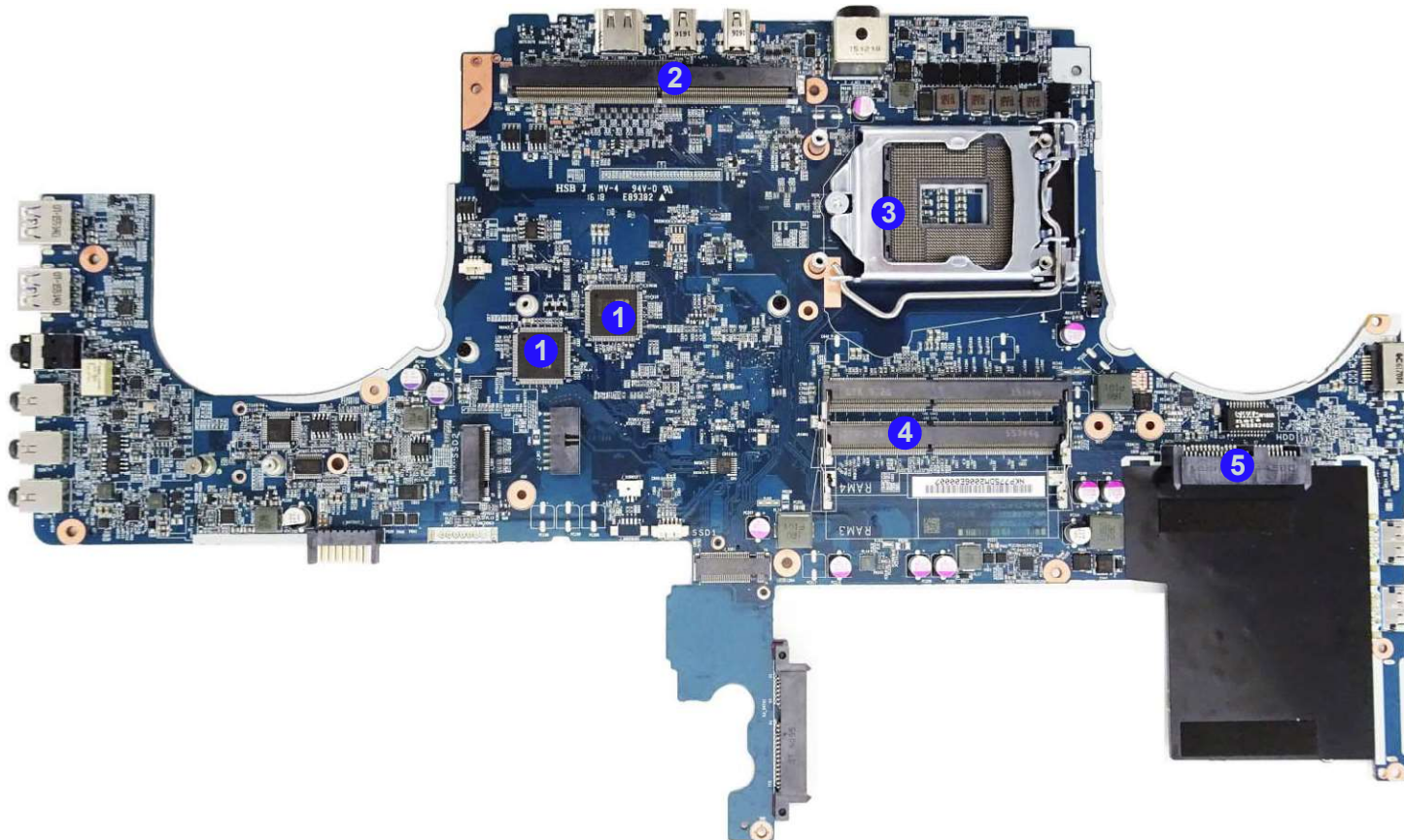
Mainboard Overview - Top (Key Parts)



Mainboard Overview - Bottom (Key Parts)

Figure 8
**Mainboard Bottom
Key Parts**

1. KBC ITE IT8587
2. VGA-Card Connector
3. CPU Socket (no CPU installed)
4. Memory Slots DDR4 SO-DIMM (Primary)
5. Hard Disk Connector

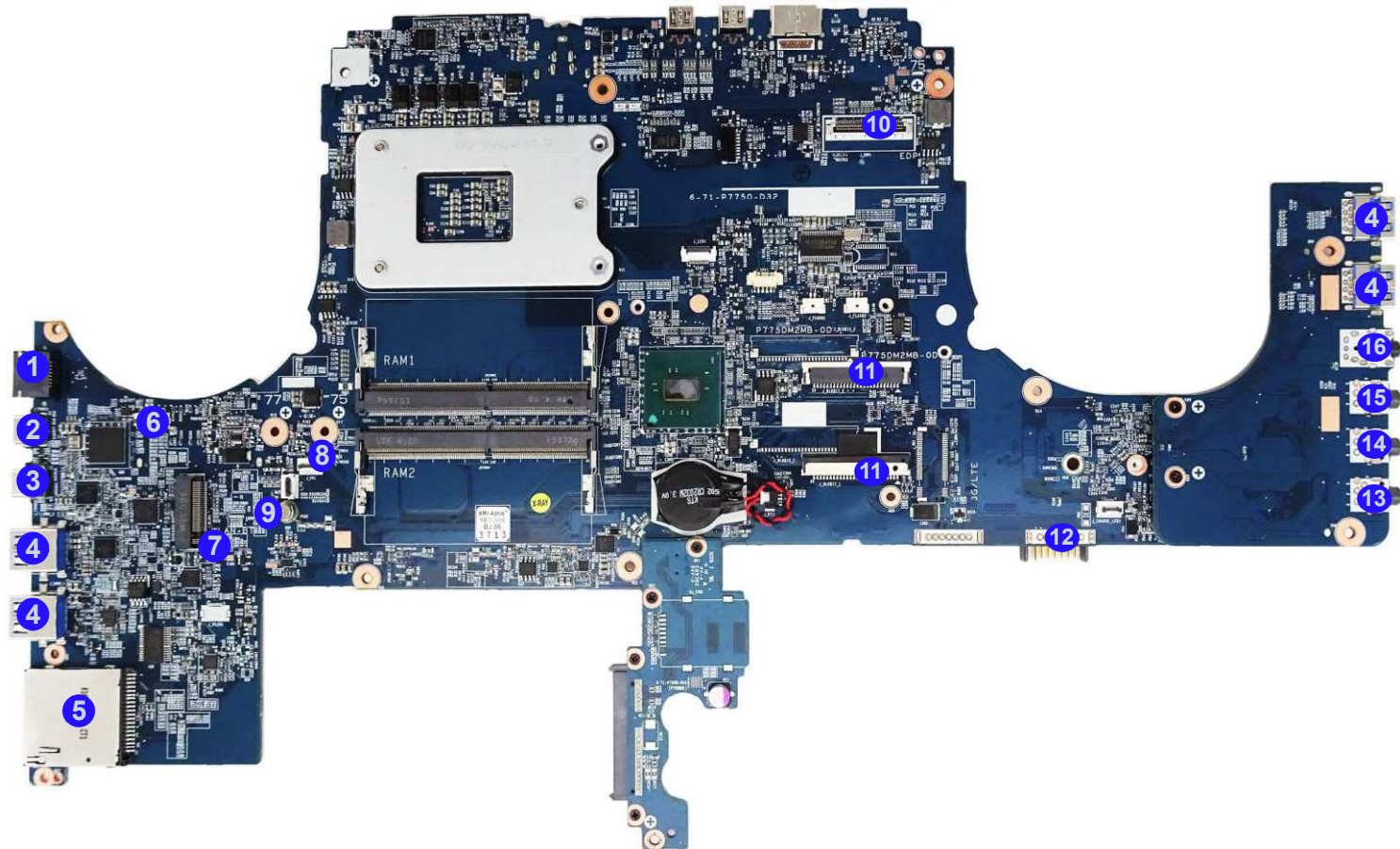


Introduction

Figure 9
**Mainboard Top
Connectors**

Mainboard Overview - Top (Connectors)

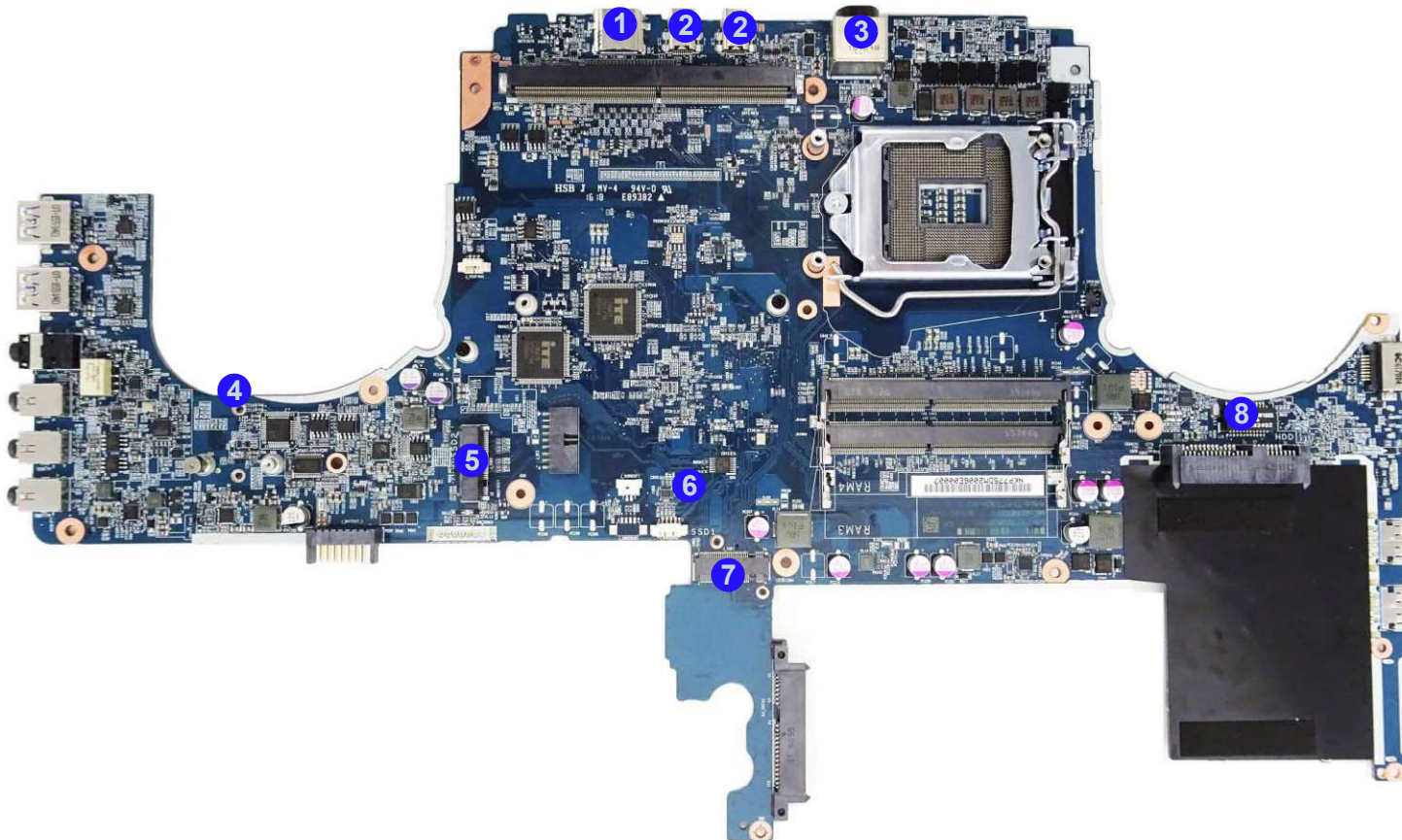
1. RJ-45 LAN Jack
2. USB 3.1/
Thunderbolt
Combo Port
3. USB 3.1 Port
4. USB 3.0/3.1 Port
5. Multi-in-1 Card
Reader
6. KB LED
Connector
7. WLAN Card
Connector
8. Fingerprint
Connector
9. TP FFC Cable
Connector
10. Panel Cable
Connector
11. Keyboard Cable
Connector
12. Battery
Connector
13. Line-In Jack
14. Microphone Jack
15. Line-Out Jack
16. Headphone and
S/PDIF Combo
Jack



Mainboard Overview - Bottom (Connectors)

Figure 10
**Mainboard Bottom
Connectors**

1. HDMI-Out Port
2. Display Port
3. DC-In Jack
4. VGA Fan Cable Connector
5. 3G / SSD Connector
6. CMOS Battery
7. SSD Connector
8. CPU Fan Cable Connector



Chapter 2: Disassembly

Overview

This chapter provides step-by-step instructions for disassembling the *P775TM (-G) / P775TM1 (-G)* series notebook's parts and subsystems. When it comes to reassembly, reverse the procedures (unless otherwise indicated).

We suggest you completely review any procedure before you take the computer apart.

Procedures such as upgrading/replacing the RAM, optical device and hard disk are included in the User's Manual but are repeated here for your convenience.

To make the disassembly process easier each section may have a box in the page margin. Information contained under the figure # will give a synopsis of the sequence of procedures involved in the disassembly procedure. A box with a  lists the relevant parts you will have after the disassembly process is complete. **Note:** The parts listed will be for the disassembly procedure listed ONLY, and not any previous disassembly step(s) required. Refer to the part list for the previous disassembly procedure. The amount of screws you should be left with will be listed here also.

A box with a  will also provide any possible helpful information. A box with a  contains warnings.

An example of these types of boxes are shown in the sidebar.


Information

Warning

Disassembly

NOTE: All disassembly procedures assume that the system is turned **OFF**, and disconnected from any power supply (the battery is removed too).

Maintenance Tools

The following tools are recommended when working on the notebook PC:

- M3 Philips-head screwdriver
- M2.5 Philips-head screwdriver (magnetized)
- M2 Philips-head screwdriver
- Small flat-head screwdriver
- Pair of needle-nose pliers
- Anti-static wrist-strap



Connections

Connections within the computer are one of four types:

Locking collar sockets for ribbon connectors

To release these connectors, use a small flat-head screwdriver to gently pry the locking collar away from its base. When replacing the connection, make sure the connector is oriented in the same way. The pin1 side is usually not indicated.

Pressure sockets for multi-wire connectors

To release this connector type, grasp it at its head and gently rock it from side to side as you pull it out. Do not pull on the wires themselves. When replacing the connection, do not try to force it. The socket only fits one way.

Pressure sockets for ribbon connectors

To release these connectors, use a small pair of needle-nose pliers to gently lift the connector away from its socket. When replacing the connection, make sure the connector is oriented in the same way. The pin1 side is usually not indicated.

Board-to-board or multi-pin sockets

To separate the boards, gently rock them from side to side as you pull them apart. If the connection is very tight, use a small flat-head screwdriver - use just enough force to start.

Maintenance Precautions

The following precautions are a reminder. To avoid personal injury or damage to the computer while performing a removal and/or replacement job, take the following precautions:

1. **Don't drop it.** Perform your repairs and/or upgrades on a stable surface. If the computer falls, the case and other components could be damaged.
2. **Don't overheat it.** Note the proximity of any heating elements. Keep the computer out of direct sunlight.
3. **Avoid interference.** Note the proximity of any high capacity transformers, electric motors, and other strong magnetic fields. These can hinder proper performance and damage components and/or data. You should also monitor the position of magnetized tools (i.e. screwdrivers).
4. **Keep it dry.** This is an electrical appliance. If water or any other liquid gets into it, the computer could be badly damaged.
5. **Be careful with power.** Avoid accidental shocks, discharges or explosions.
 - Before removing or servicing any part from the computer, turn the computer off and detach any power supplies.
 - When you want to unplug the power cord or any cable/wire, be sure to disconnect it by the plug head. Do not pull on the wire.
6. **Peripherals** – Turn off and detach any peripherals.
7. **Beware of static discharge.** ICs, such as the CPU and main support chips, are vulnerable to static electricity. Before handling any part in the computer, discharge any static electricity inside the computer. When handling a printed circuit board, do not use gloves or other materials which allow static electricity buildup. We suggest that you use an anti-static wrist strap instead.
8. **Beware of corrosion.** As you perform your job, avoid touching any connector leads. Even the cleanest hands produce oils which can attract corrosive elements.
9. **Keep your work environment clean.** Tobacco smoke, dust or other air-born particulate matter is often attracted to charged surfaces, reducing performance.
10. **Keep track of the components.** When removing or replacing any part, be careful not to leave small parts, such as screws, loose inside the computer.

Cleaning

Do not apply cleaner directly to the computer, use a soft clean cloth.

Do not use volatile (petroleum distillates) or abrasive cleaners on any part of the computer.

(For Computer Models Supplied with Light Blue Cleaning Cloth) Some computer models in this series come supplied with a light blue cleaning cloth. To clean the computer case with this cloth follow the instructions below.

- Power off the computer and peripherals.
- Disconnect the AC/DC adapter from the computer.
- Use a little water to dampen the cloth slightly.
- Clean the computer case with the cloth.
- Dry the computer with a dry cloth, or allow it time to dry before turning on.
- Reconnect the AC/DC adapter and turn the computer on.



Power Safety Warning

Before you undertake any upgrade procedures, make sure that you have turned off the power, and disconnected all peripherals and cables (including telephone lines and power cord). It is advisable to also remove your battery in order to prevent accidentally turning the machine on.

Disassembly Steps

The following table lists the disassembly steps, and on which page to find the related information. **PLEASE PERFORM THE DISASSEMBLY STEPS IN THE ORDER INDICATED.**

To remove the Battery:

1. Remove the battery [page 2 - 5](#)

To remove the HDD:

1. Remove the battery [page 2 - 5](#)
2. Remove the HDD [page 2 - 6](#)

To remove the M.2 SSD:

1. Remove the battery [page 2 - 5](#)
2. Remove the HDD [page 2 - 6](#)
3. Remove the M.2 SSD [page 2 - 9](#)

To remove the Primary System Memory:

1. Remove the battery [page 2 - 5](#)
2. Remove the primary system memory [page 2 - 10](#)

To remove and install the Processor:

1. Remove the battery [page 2 - 5](#)
2. Remove the system memory [page 2 - 10](#)
3. Remove the processor [page 2 - 12](#)
4. Install the processor [page 2 - 14](#)

To remove the System Memory under the Keyboard:

1. Remove the battery [page 2 - 5](#)
2. Remove the processor [page 2 - 12](#)
3. Remove the keyboard [page 2 - 15](#)
4. Remove the system memory [page 2 - 16](#)

To remove the WLAN Module:

1. Remove the battery [page 2 - 5](#)
2. Remove the processor [page 2 - 12](#)
3. Remove the keyboard [page 2 - 10](#)
4. Remove the wireless LAN [page 2 - 17](#)

To remove the WiGig Module:

1. Remove the battery [page 2 - 5](#)
2. Remove the processor [page 2 - 12](#)
3. Remove the keyboard [page 2 - 10](#)
4. Remove the WiGig [page 2 - 19](#)

To remove and install the M.2 SSD:

1. Remove the battery [page 2 - 5](#)
2. Remove the primary system memory [page 2 - 10](#)
3. Remove the M.2 SATA [page 2 - 20](#)
4. Install the M.2 SATA [page 2 - 21](#)

To remove and install the Video Card:

1. Remove the battery [page 2 - 5](#)
2. Remove the video card [page 2 - 22](#)
3. Install the video card [page 2 - 23](#)

Removing the Battery

1. Turn the computer **off**, and turn it over.
2. Slide the latch **1** in the direction of the arrow (*Figure 1a*).
3. Slide the latch **2** in the direction of the arrow, and hold it in place (*Figure 1a*).
4. Lift the battery in the direction of the arrow **3**.
5. Lift the battery **4** out of the compartment (*Figure 1c*).

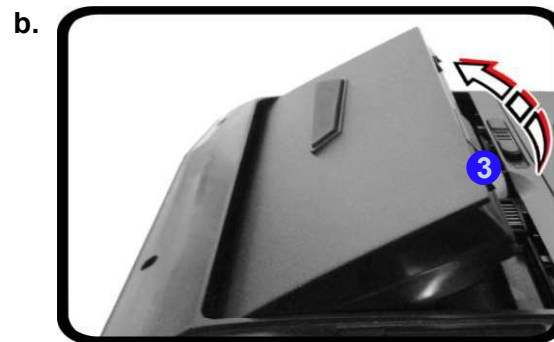
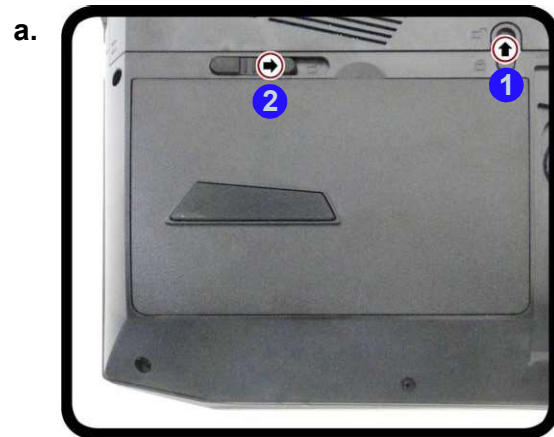
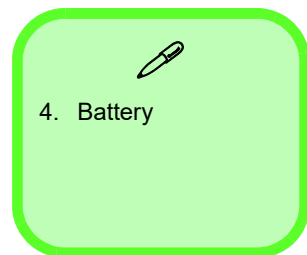


Figure 1
Battery Removal

- a. Slide the latch and hold in place.
- b. Lift the battery up toward the direction of the arrow.
- c. Lift the battery out.



Disassembly

Figure 2
**HDD Assembly
Removal**

- Locate the HDD bay cover and remove the screws.
- Remove the hard disk bay cover by sliding the cover at point ③.

Removing and Installing the Hard Disk Drive

The hard disk drive can be taken out to accommodate other 2.5" serial (SATA) hard disk drives with a height of 7mm/ 9.5mm (h). Follow your operating system's installation instructions, and install all necessary drivers and utilities (as outlined in **Chapter 4 of the User's Manual**) when setting up a new hard disk.

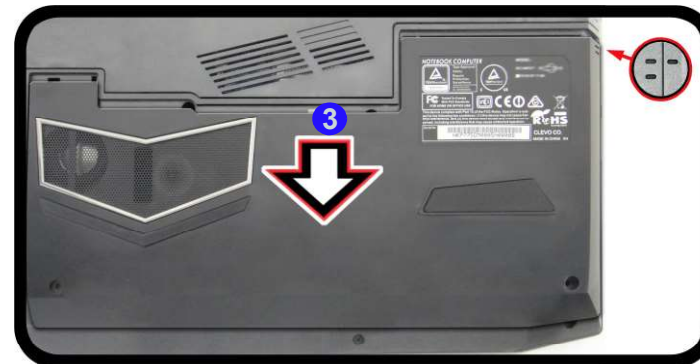
Hard Disk Removal Process

- Turn **off** the computer, and remove the battery ([page 2 - 5](#)).
- Locate the hard disk bay cover and remove screws ① - ② ([Figure 2a](#)).
- Remove the hard disk bay cover by sliding the cover at point ③ ([Figure 2b](#)).

a.



b.



- 2 Screws



HDD System Warning

New HDD's are blank. Before you begin make sure:

You have backed up any data you want to keep from your old HDD.

You have all the CD-ROMs and FDDs required to install your operating system and programs.

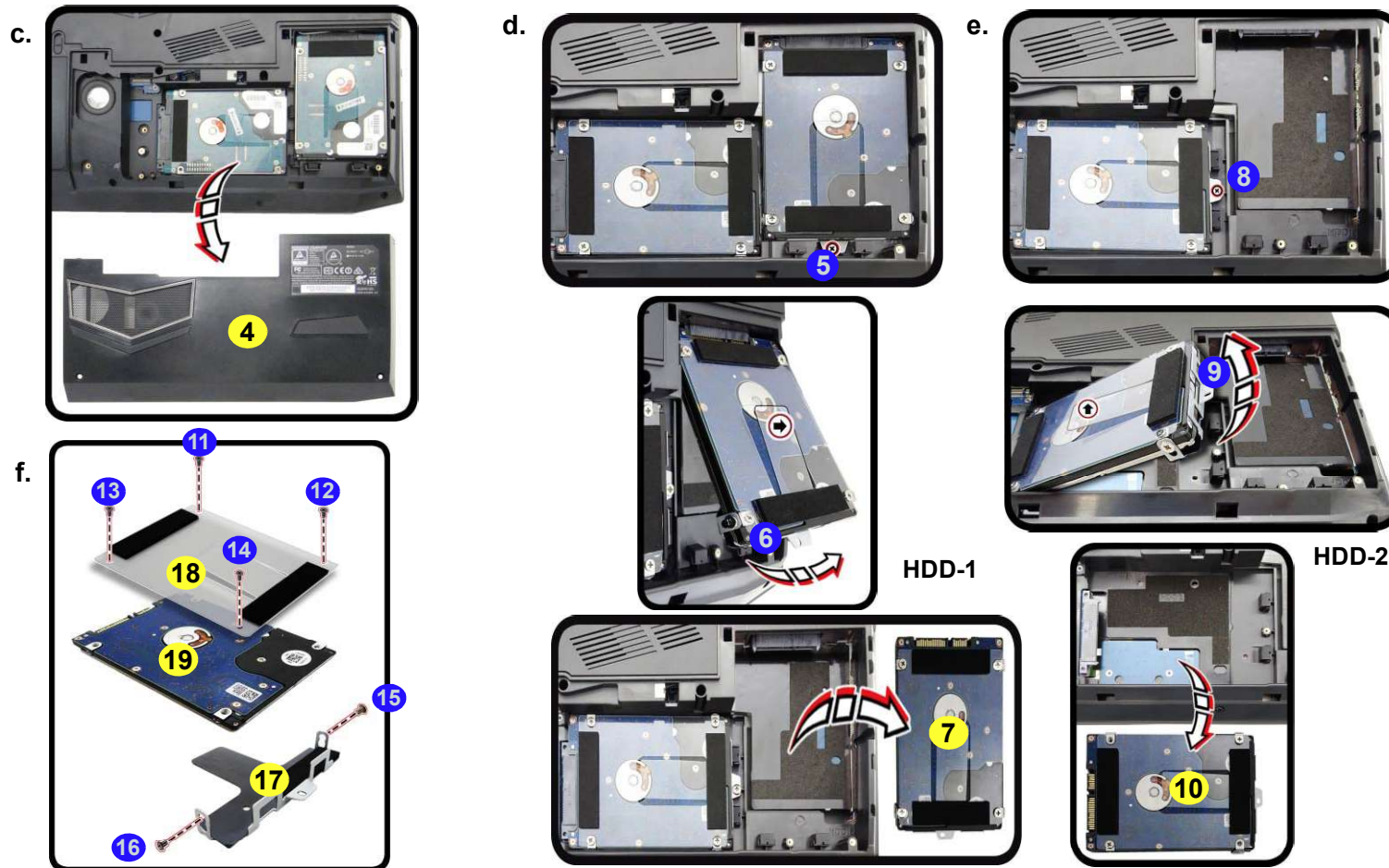
If you have access to the internet, download the latest application and hardware driver updates for the operating system you plan to install. Copy these to a removable medium.

4. Lift the hard disk bay cover **4** off the computer (**Figure 3c**)
5. Remove the screw **5**. Slightly lift and pull the HDD-1 assembly in the direction of the arrow **6** to remove the hard disk assembly **7** (**Figure 3d**).
6. Remove the screw **8**. Slightly lift and pull the HDD-2 assembly (if available) in the direction of the arrow **9** to remove the hard disk assembly **10** (**Figure 3e**).
7. Remove screws **11** - **16**, HDD bracket **17** and the adhesive cover **18** from the hard disk **19** (**Figure 3f**).
8. Reverse the process to install a new hard disk (do not forget to replace all the screws and covers).

Figure 3

HDD Assembly Removal (cont'd.)

- c. Remove the HDD bay cover.
- d. Remove the screw. Lift and pull the HDD-1 assembly in the direction of the arrow to remove the hard disk assembly.
- e. Remove the screw. Lift and pull the HDD-2 assembly in the direction of the arrow to remove the hard disk assembly.
- f. Remove the screws, hdd bracket and adhesive cover.

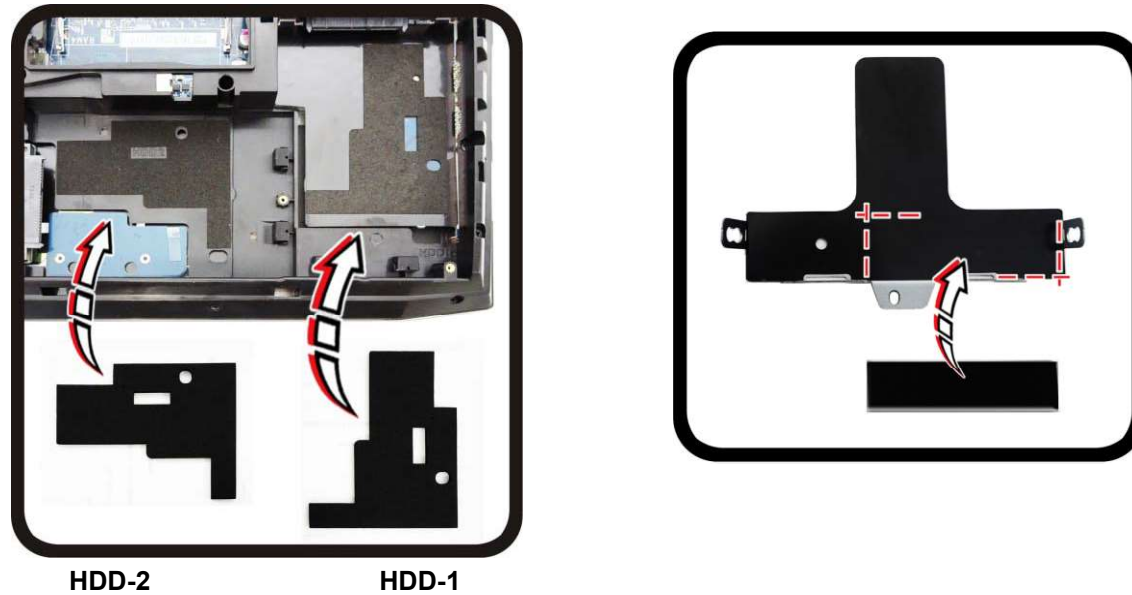


Disassembly

Hard Disk Size Note (Foam Rubber Insert)

Note that the hard disks pictured on the following pages are all 9.5mm(H) hard disk drives. In some cases 7mm(H) hard disk drives will be installed.

Figure 4
**Foam Rubber
Insert for 7mm(H)
HDDs**



- If you are replacing a 9.5mm(H) HDD with a 7mm(H) HDD then insert the foam rubber insert.
- If you are replacing a 7mm(H) HDD with a 9.5mm(H) HDD then remove the foam rubber insert.

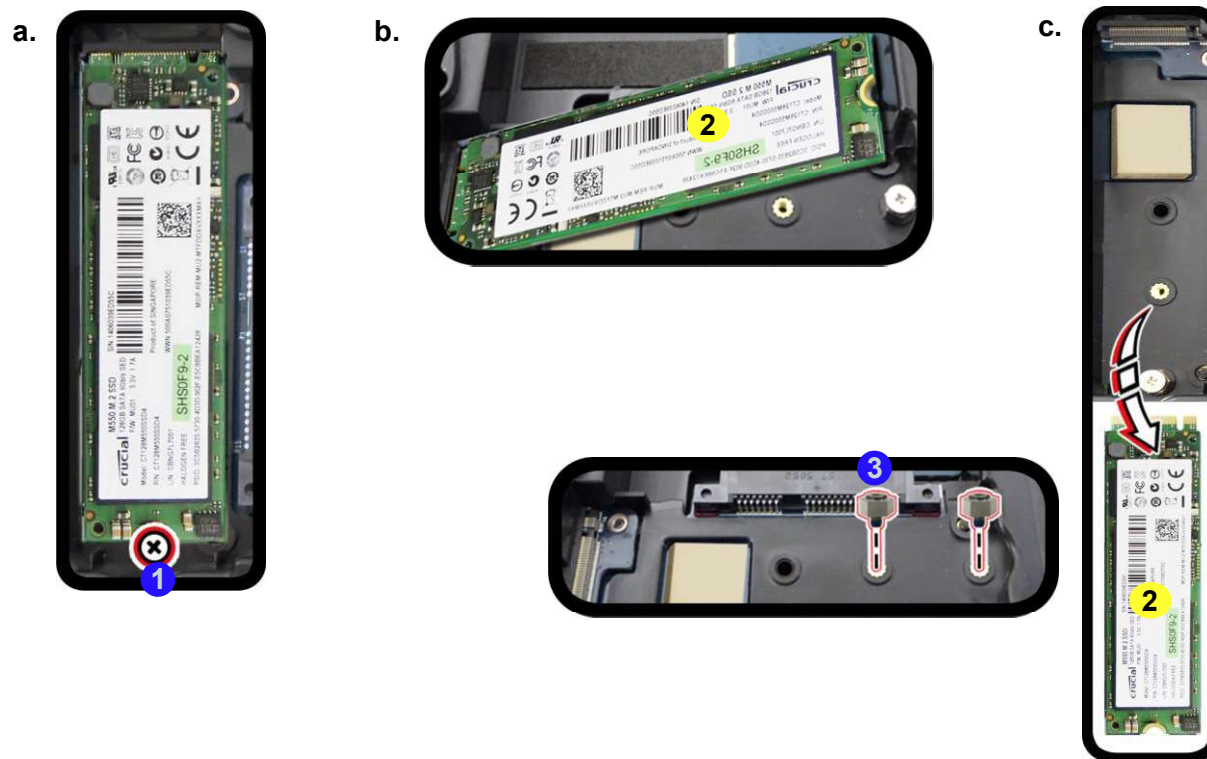
Removing the M.2 SSD Module

Note that the SSD (if installed) is beside the HDD bay.

1. Turn **off** the computer, and turn it over, remove the battery ([page 2 - 5](#)).
2. Remove the screw **1** from the SSD ([Figure 5a](#)).
3. The M.2 SSD module **2** will pop-up ([Figure 5b](#)).
4. Lift the M.2 SSD module **2** up and off the computer ([Figure 5c](#)).
5. Reverse the process to install a new SSD (make sure that the hexagonal screw **3** is in the correct location depending upon the size of the module).

Figure 5
M.2 SSD Module Removal

- a. Remove the screws.
- b. The module will pop up.
- c. Lift the module out.



- 
2. M.2 SSD Module
- 1 Screw

Disassembly

Figure 6
RAM Module Removal

- a. Remove the screws. Slide the bottom cover until the cover and case indicators are aligned.

Removing the Primary System Memory (RAM)

The computer has **four** memory sockets for 260 pin Small Outline Dual In-line (SO-DIMM) **DDR 4** type memory modules. The total memory size is automatically detected by the POST routine once you turn on your computer.

Two primary memory sockets are located under component bay cover (the bottom case cover), and two secondary memory sockets are located under the keyboard (not user upgradable). If you are installing only two RAM modules then they should be installed in the primary memory sockets under the component bay cover.

Note that the RAM located under the keyboard is not user upgradable.

Memory Upgrade Process

1. Turn **off** the computer, and turn it over, remove the battery ([page 2 - 5](#)).
2. Remove screws **1** - **5**.
3. Slide the bottom cover until the cover and case indicators **6** are aligned ([Figure 6a](#)).

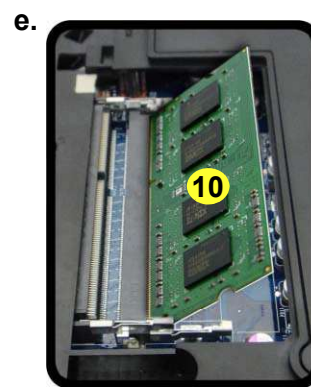
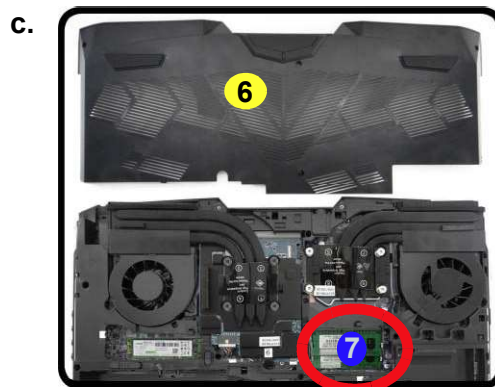


- 5 Screws

Figure 7
RAM Module Removal (cont'd.)

- c. Lift the component bay cover off the computer case. The modules will be visible at point 7.
- d. Gently pull the two release latches on the sides of the memory socket(s) in the direction indicated below.
- e. The RAM module will pop-up, and you can remove it.

4. Lift the component bay cover 6 off the computer case. The modules will be visible at point 7 (Figure 7c).
5. Gently pull the two release latches (8 & 9) on the sides of the memory socket(s) in the direction indicated below (Figure 7d).
6. The RAM module 10 will pop-up, and you can remove it (Figure 7e).
7. Pull the latches to release the second module if necessary.
8. Insert a new module holding it at about a 30° angle and fit the connectors firmly into the memory slot.
9. The module's pin alignment will allow it to only fit one way. Make sure the module is seated as far into the slot as it will go. DO NOT FORCE the module; it should fit without much pressure.
10. Press the module in and down towards the mainboard until the slot levers click into place to secure the module.
11. Replace the bay cover and screws.
12. Restart the computer to allow the BIOS to register the new memory configuration as it starts up.



Contact Warning

Be careful not to touch the metal pins on the module's connecting edge. Even the cleanest hands have oils which can attract particles, and degrade the module's performance.



- 6. Component Bay Cover
- 10. RAM Module

- 4 Screws

Disassembly

Figure 8
Processor Removal Procedure

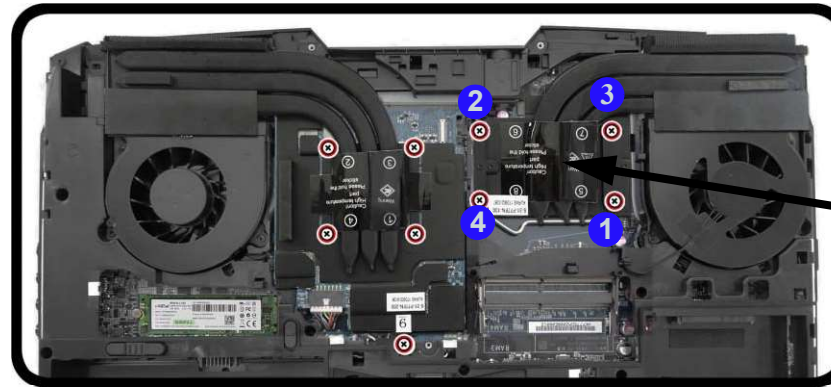
- a. Remove the screws in the correct order.
- b. Carefully remove the heat sink unit.

Removing and Installing the Processor

Processor Removal Procedure

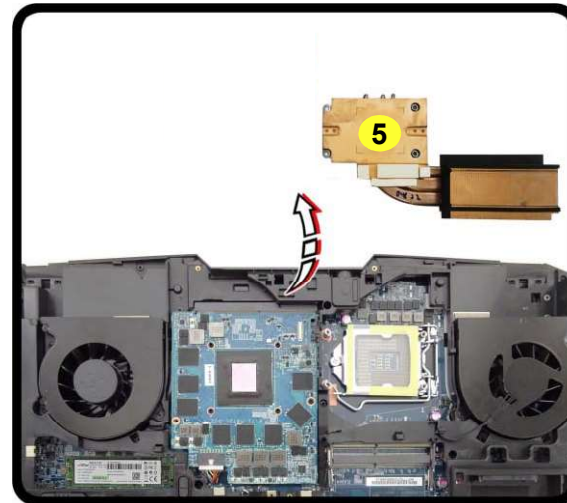
1. Turn **off** the computer, remove the battery ([page 2 - 5](#)), and component bay cover ([page 2 - 10](#)).
2. Remove screws **1** - **4** from the heat sink unit in the order indicated on the label (i.e screw **4** first through to screw **1** last [Figure 8a](#)).
3. Carefully (it may be hot) remove the heat sink unit **5** ([Figure 8b](#)).

a.



Note:
Loosen the screws in the reverse order 4-3-2-1 as indicated.

b.



5. Heat Sink Unit

- 4 Screws

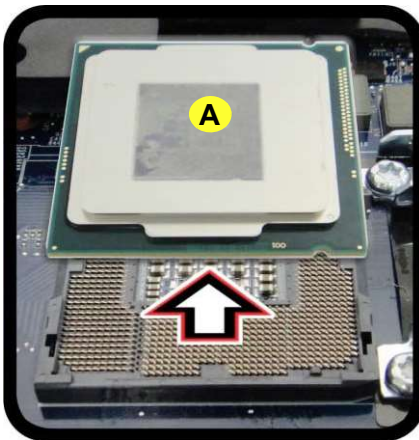
4. Press down and hold the latch **11** (with the latch held down you will be able to release it).
5. Move the latch **11** and bracket **12** fully in the direction indicated to unlock the CPU (**Figure 9c**).
6. Carefully (it may be hot) lift the CPU **A** up out of the socket (**Figure 9d**).
7. See **page 2 - 14** for information on inserting a new CPU.
8. When re-inserting the CPU, pay careful attention to the pin alignment, it will fit only one way (DO NOT FORCE IT!).

c.



Unlock

d.



Caution

The heat sink, and CPU area in general, contains parts which are subject to high temperatures. Allow the area time to cool before removing these parts.

Figure 9

Processor Removal (cont'd)

- c. Move the latch and bracket fully in the direction indicated to unlock the CPU.
- d. Lift the CPU out of the socket.



A. CPU

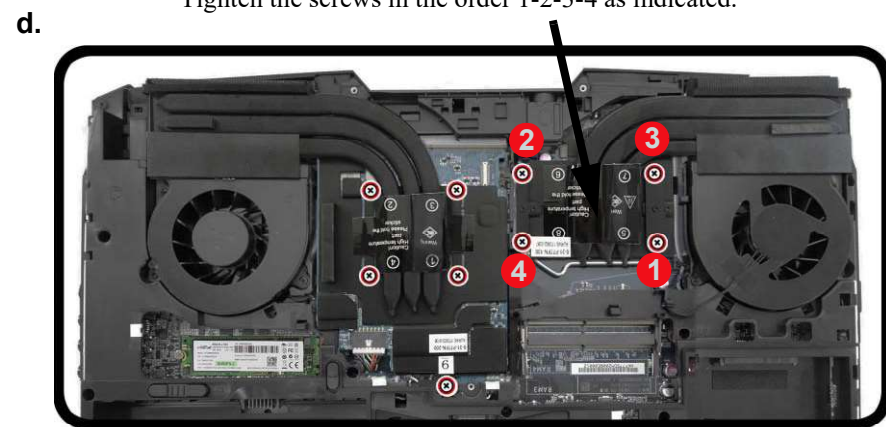
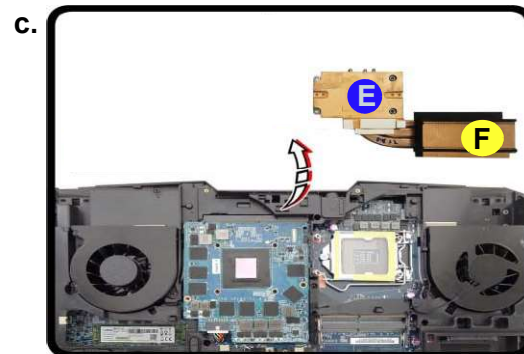
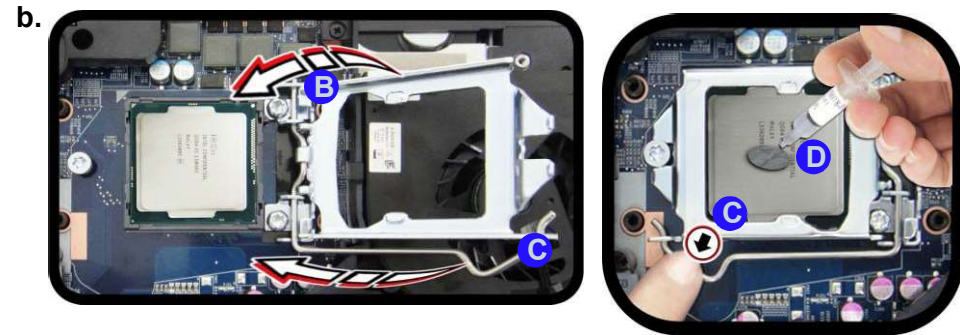
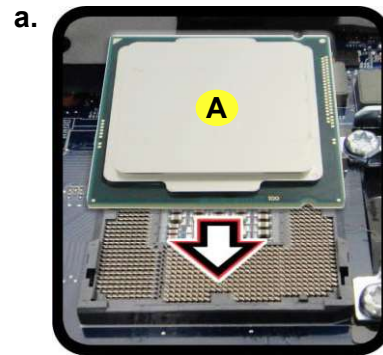
Disassembly

Figure 10
Processor Installation

- Insert the CPU.
- Move the latch and bracket fully in the direction indicated to lock the CPU. Apply thermal grease.
- Remove the sticker from the heat sink unit and insert the heat sink.
- Tighten the screws.

Processor Installation Procedure

- Insert the CPU **A**; pay careful attention to the pin alignment (*Figure 10a*), it will fit only one way (DO NOT FORCE IT!).
- Move the bracket **B** and latch **C** fully in the direction indicated to lock the CPU.
- Apply the whole tube of thermal grease **D** to the top of the CPU as shown (*Figure 10b*).
- Remove the sticker **E**** (*Figure 10c*) from the heat sink unit (if it is a new unit).
- Insert the heat sink unit **F** as indicated in *Figure 10c*.
- Tighten the CPU heat sink screws in the order **1** - **4** (the order as indicated on the label and *Figure 10d*).
- Replace the CPU fan, component bay cover and tighten the screws (*page 2 - 12*).



Note:

Tighten the screws in the order 1-2-3-4 as indicated.

- A. CPU
F. Heat Sink
- 4 Screws

Removing the System Memory (RAM) from Under the Keyboard

The computer has **four** memory sockets for 260 pin Small Outline Dual In-line (SO-DIMM) **DDR 4** type memory modules. The total memory size is automatically detected by the POST routine once you turn on your computer.

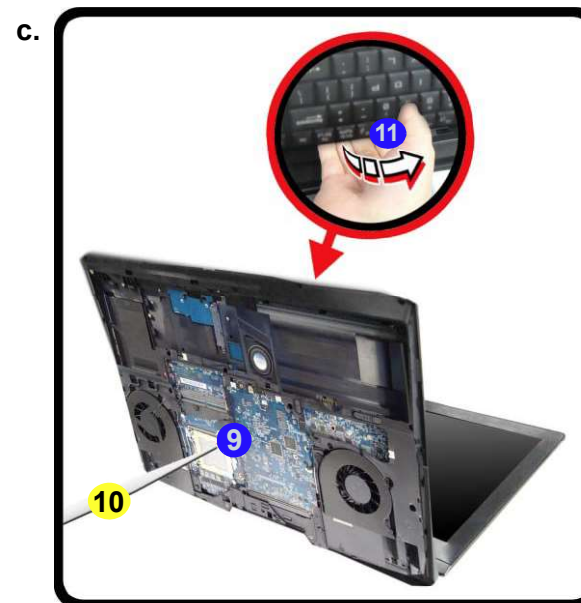
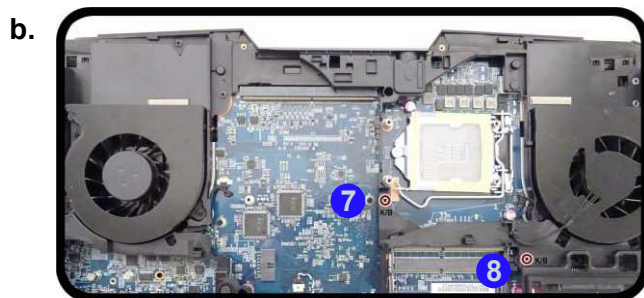
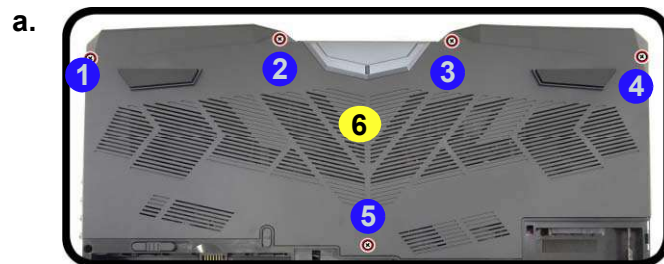
Two primary memory sockets are located under component bay cover (the bottom case cover), and two secondary memory sockets are located under the keyboard. If you are installing only two RAM modules then they should be installed in the primary memory sockets under the component bay cover.

Memory Upgrade Process

1. Turn **off** the computer, and turn it over, remove the battery ([page 2 - 5](#)).
2. Remove screws **1** - **5** and the component bay cover **6** ([Figure 11a](#)) and CPU heatsink ([page 2 - 12](#)).
3. Remove screws **7** - **8** from the bottom of the computer ([Figure 11b](#)).
4. Open it up with the LCD on a flat surface before pressing at point **9** to release the keyboard module (use an eject stick **10** to do this with a diameter no bigger than 2.5mm) while releasing the keyboard in the direction of the arrow **11** as shown ([Figure 11c](#)).

Figure 11
Keyboard Removal

- a. Remove the screws and component bay cover.
- b. Remove the screws.
- c. Eject the keyboard using a special eject stick to push the keyboard out while releasing the keyboard as shown.



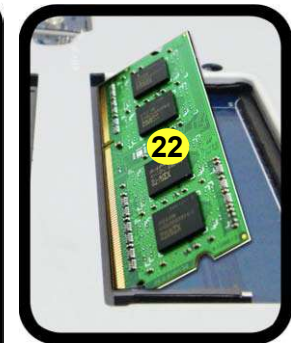
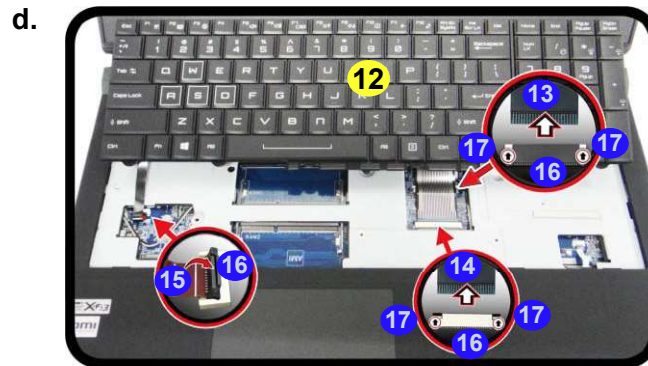
- 
6. Top Cover Module
10. Eject Stick
- 7 Screws

Disassembly

Figure 12
RAM Module Removal

- d. Lift the keyboard up, and disconnect the keyboard ribbon cable from the locking collar socket.
- e. Remove the keyboard and the memory sockets will be visible.
- f. Pull the two release latches on the sides of the memory socket(s) in the direction indicated.

5. Carefully lift the keyboard **12** up, being careful not to bend the keyboard ribbon cables **13** - **15**.
6. Disconnect the keyboard ribbon cables **13** - **15** from the locking collar socket **16** by using a small flat-head screwdriver to pry the locking collar pins **17** away from the base (*Figure 12d*).
7. Remove the keyboard and the memory sockets **18** & **19** will be visible.
8. Gently pull the two release latches (**20** & **21**) on the sides of the memory socket(s) in the direction indicated below.
9. The RAM module **22** will pop-up, and you can remove it.
10. Pull the latches to release the second module if necessary.
11. Insert a new module holding it at about a 30° angle and fit the connectors firmly into the memory slot.
12. The module's pin alignment will allow it to only fit one way. Make sure the module is seated as far into the slot as it will go. DO NOT FORCE the module; it should fit without much pressure.
13. Press the module in and down towards the mainboard until the slot levers click into place to secure the module.
14. Replace the bay cover and screws.
15. Restart the computer to allow the BIOS to register the new memory configuration as it starts up.



12. Keyboard
22. RAM Modules



Contact Warning

Be careful not to touch the metal pins on the module's connecting edge. Even the cleanest hands have oils which can attract particles, and degrade the module's performance.

Removing the Wireless LAN Module

1. Turn **off** the computer, remove the battery ([page 2 - 5](#)), CPU ([page 2 - 12](#)) and the keyboard ([page 2 - 15](#)).
2. The Wireless LAN module will be visible at point **1** under the keyboard ([Figure 13a](#)).
3. Carefully disconnect cables **2** - **3**, then remove screw **4** from the module socket ([Figure 13b](#)).
4. The Wireless LAN module **5** will pop-up ([Figure 13c](#)).
5. Lift the Wireless LAN module ([Figure 13d](#)) up and off the computer.

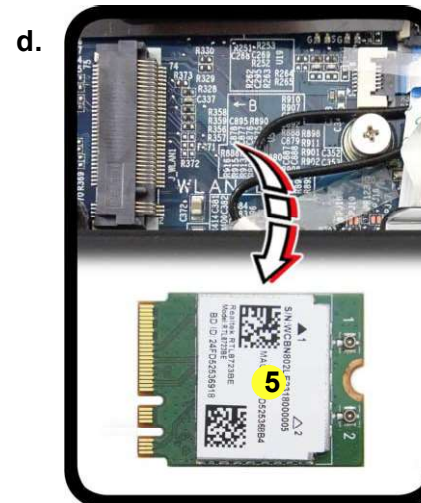
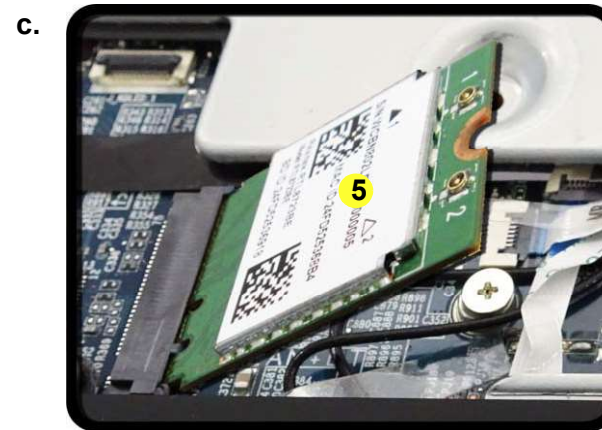
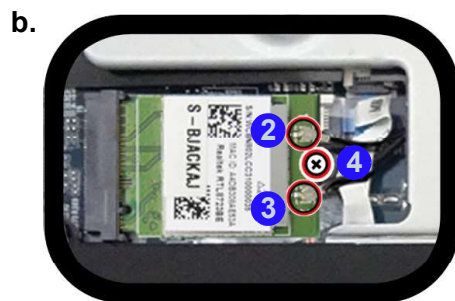


Figure 13
**Wireless LAN
Module Removal**

- a. The Wireless LAN module will be visible at point **1** under the keyboard
- b. Disconnect the cables and remove the screw.
- c. The WLAN module will pop up.
- d. Lift the WLAN module out.



5. WLAN Module

- 1 Screw

Wireless LAN, Combo Module Cables

Note that the cables for connecting to the antennae on WLAN, WLAN & Bluetooth Combo, 3G and LTE modules are not labelled. The cables/covers (each cable will have either a black or transparent cable cover) are color coded for identification as outlined in the table below.

Module Type	Antenna Type	Cable Color	Cable Cover Type
WLAN/WLAN & Bluetooth Combo	WM 1	Black	Transparent
	WM 2	Black	White
WiGig	WG 1	Blue	Black

Cable 1 is usually connected to antenna 1 (Main) on the module, and cable 2 to antenna 2 (Aux).

Removing the WiGig Module

1. Turn **off** the computer, remove the battery ([page 2 - 5](#)), CPU ([page 2 - 12](#)) and the keyboard ([page 2 - 15](#)).
2. The module will be visible at point **1** under the keyboard ([Figure 13a](#)).
3. Carefully disconnect cables **2** - **4**, then remove screw **5** from the module socket ([Figure 13b](#)).
4. The module **6** will pop-up ([Figure 13c](#)).
5. Lift the module ([Figure 13d](#)) up and off the computer.

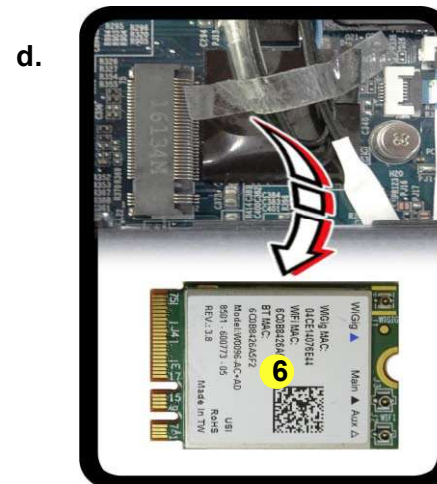
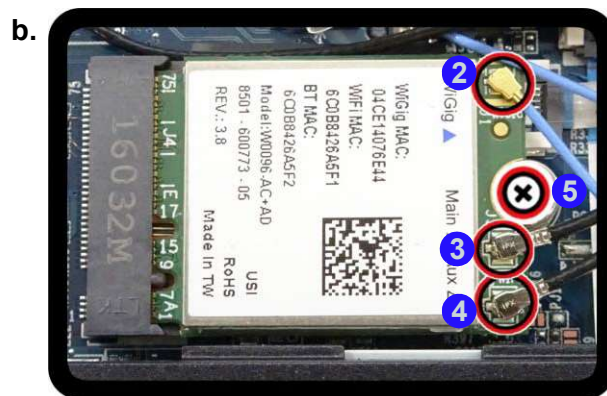


Figure 14
WiGig Module
Removal

- a. The module will be visible at point **1** under the keyboard
- b. Disconnect the cables and remove the screw.
- c. The module will pop up.
- d. Lift the module out.



6. WiGig Module

- 1 Screw

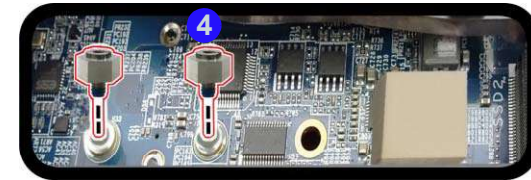
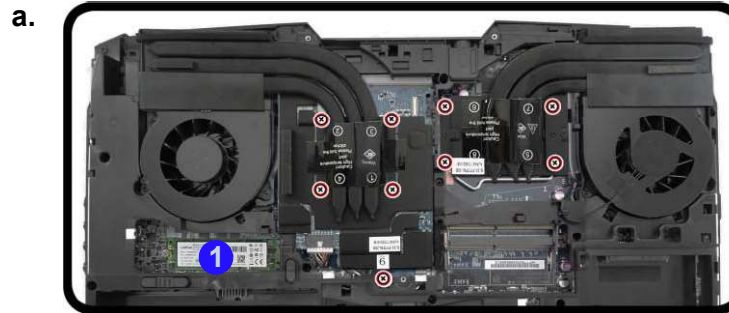
Disassembly

Figure 15
M.2 SSD Module Removal

- Locate the module.
- Remove the screw.
- The module will pop-up.
- Lift the module up off the socket.

Removing the M.2 SSD Module

- Turn **off** the computer, remove the battery ([page 2 - 5](#)), and component bay cover ([page 2 - 10](#)).
- Locate the module; it is visible at point **1** ([Figure 15a](#)).
- Carefully remove the screw **2** from the module ([Figure 15b](#)).
- The M.2 SATA module **3** will pop-up ([Figure 15c](#)).
- Lift the M.2 SATA module **3** up and off the computer ([Figure 15d](#)).
- Reverse the process to install a new SSD (make sure that the hexagonal screw **4** is in the correct location).



3. SSD Module

- 1 Screw

M.2 SSD Installation Procedure

1. Place the thermal pad **1** on the computer as shown (**Figure 16a**).
2. Insert the module **2** in the computer. Make sure that the hexagonal screw **3** is in the correct location (**Figure 16b**).
3. Tighten the screw **4** to secure it in place (**Figure 16c**).

Figure 16
M.2 SSD Module Installation

- a. Place the thermal pad.
- b. Insert the module.
- c. Tighten the screw.

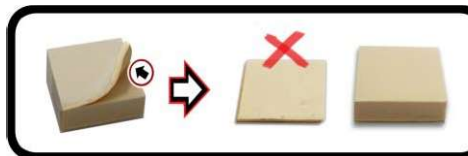
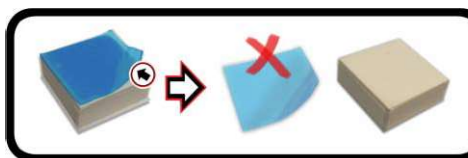


Thermal Pad

Make sure you place the thermal pad's adhesive side down onto the computer surface as illustrated.

The usage of the thermal pad will depend upon the thickness of the module being used.

- If you are using the thinner module, then apply the whole thermal pad provided on the computer.
- If you are using the thicker module, separate the pad into its two parts. Use the larger part and place the adhesive side onto the computer (discard the smaller part that you have separated).



1. Thermal Pad
2. M.2 SATA Module

- 1 Screw